

연세대학교
카피킬러캠퍼스 AI생성검사
요약 결과확인서

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성 명
서 명

이메일	2025324176	AI작성률	9%
성명	자필로 기재하세요		
소속	의료기기산업학과		
검사번호	00369780028		
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검사 유형	✔ AI생성검사	발급형태	▼ 기본보기 ✔ 요약보기 ▼ 상세보기
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비고			

분석 정보					
AI작성률	전체 영역	AI 의심 영역	높음	중간	낮음
9%	24	2	0	0	2

본 결과확인서는 2025324176 사용자가 카피킬러에서 수행한 문서 검사에 대한 분석 결과로 카피킬러 시스템이 생성한 자료입니다. 문서 작성과 평가에 대한 기준이 각 학교, 기관마다 다르므로 본 분석 결과와 최종 평가자의 평가는 다를 수 있습니다.

낮음

To construct the fixed-length AEC-128 feature, the axial source series was first identified for each patient, excluding coronal/sagittal reformations, scout images, and dose reports, on the basis of slice count, DICOM Image Type, Series Description, and image orientation. An open-source, deep-learning-based segmentation tool, TotalSegmentator [4], was then applied to localize the liver and the bilateral hip bones, with the cranial-most liver slice and caudal-most hip slice defining the upper and lower boundaries of the analysis range; segmentation was used solely for this anatomical landmark localization, while tube-current values themselves were obtained directly from the DICOM header field X-Ray Tube Current. Tube-current values within this hip-bone-to-liver range were extracted in craniocaudal order and, because the resulting slice count varied across patients, resampled by linear interpolation to a fixed length of 128 points, yielding the AEC-128 curve used as the analytic unit in subsequent modeling.

① AI생성 특징 기계적 정확성 및 균일성

낮음

The mean curve $\mu(z)$, for $z = 1, \dots, 128$, was first computed across all curves in the internal cohort, and each patient's deviation curve was obtained by subtracting this mean from the raw curve, $d_i(z) = x_i(z) - \mu(z)$. The 128×128 covariance matrix of the deviation curves was eigen-decomposed via singular valuedecomposition [7], using scikit-learn's PCA implementation for numerical stability, yielding orthonormal eigenfunctions φ_k and their associated eigenvalues λ_k , ranked by the proportion of variance explained. Each patient's deviation curve was then projected onto each retained eigenfunction to obtain the componentscore, $\text{score}_{i,k} = \sum_r d_i(z) \cdot \varphi_k(z)$ - the FPCA score used as the AEC term in the logistic regression models below.

① AI생성 특징 기계적 정확성 및 균일성